

Fig. 3: Actual-size PCB pattern of the PIR motion-sensing SSR switch

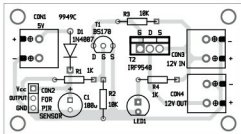


Fig. 4: Component layout of PCB

on when sensor output goes high. Output of the PIR sensor module is connected to the base of T1 (BS170) via resistor R1.

When motion is detected, PIR sensor output goes high to about 3.3V. The high-level, standard TTL output from the sensor-module switches on MOSFET T2 through T1, and T2 is closed by this action. As a result, the connected DC load is powered through T2 for a finite duration, determined by the time-delay setting of the PIR sensor module. Usually, the PIR sensor module includes an onboard time control preset pot (Fig. 1), which is adjustable from seconds to minutes. LED1 is the power-on indicator.

Construction and testing

An actual-size, single-side PCB for the PIR motion-sensing SSR switch is shown in Fig. 3 and its component layout in Fig. 4. Enclose the PCB in a small box so that you can connect 12V IN and 12V OUT easily at the rear side of

the box. Install the PIR sensor at a suitable place and connect it to the PCB using a three-wire cable.

EFY notes.

1. During initial power-up, the circuit automatically switches to active mode for a while and then shifts back to sleep mode.
2. p-channel power MOSFET IRF 9540 (in TO-220 package) is preferred universally for all commercial/industrial applications at power dissipation levels up to approximately 50 watts.
3. According to the data sheet, the PIR sensor requires an initial stabilisation time of 10 to 60 seconds in order to function properly. During this time, any motion in its field-of-view (approx. 6 metres) should be avoided. **EFY**



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